

MANIKANDAA S

Data Scientist | Machine Learning Engineer | Python • Deep Learning

Tiruchirappalli, Tamil Nadu, India | +91-9445219410 | manikandaa944@gmail.com | [LinkedIn](#) | [GitHub](#)

SUMMARY

Data Science graduate with hands-on experience building end-to-end machine learning pipelines, achieving up to 87% accuracy on healthcare prediction tasks. Skilled in EDA, feature engineering, statistical modeling, and deep learning (CNNs) using Python, Scikit-learn, TensorFlow, and Keras. Completed a government internship applying statistical inference to real-world datasets at the District Statistical Office, Trichy. Holds an M.Sc. in Data Science from Bharathidasan University and a B.Sc. in Statistics, combining strong theoretical foundations with practical model-building experience across classification, computer vision, and recommendation systems.

SKILLS

Programming: Python, R, SQL (MySQL)

Machine Learning / Deep Learning: Logistic Regression, Random Forest, CNN, TensorFlow, Keras, Scikit-learn

Data Science: EDA, Feature Engineering, Data Cleaning, Statistical Modeling, Predictive Analytics

Model Evaluation: Accuracy, ROC-AUC, Confusion Matrix, Precision, Recall, F1-Score, Cross-Validation

Visualization & BI: Power BI, Matplotlib, Seaborn

Tools: Pandas, NumPy, Jupyter Notebook, Google Colab, GitHub, SPSS

EXPERIENCE

Data Science Intern — Analytics Career Connect

Jun 2024 – Jul 2024

Remote

- Performed data cleaning, preprocessing, and exploratory data analysis (EDA) on real-world datasets using Python, Pandas, and NumPy.
- Created data visualizations using Matplotlib to identify trends, patterns, and actionable insights.
- Applied statistical analysis and basic machine learning techniques to analyze datasets and uncover meaningful patterns.
- Developed analytical insights and communicated findings clearly through visualizations and concise reports.
- Prepared analytical reports translating technical findings into stakeholder-ready summaries to support data-driven decision-making.

PROJECTS

Healthcare Prediction Model — Diabetes Detection

2024

Python · Scikit-learn · Logistic Regression · Random Forest · Feature Engineering

- Executed a complete supervised ML pipeline on the PIMA Indians Diabetes Dataset: preprocessing (missing value imputation, scaling) → EDA → feature selection → model training → hyperparameter tuning → evaluation.
- Built and compared Logistic Regression and Random Forest classifiers, achieving 87% prediction accuracy with Random Forest after tuning.
- Applied k-fold cross-validation to prevent overfitting and validate generalization, evaluated with ROC-AUC, confusion matrix, precision, recall, and F1-score.

Cat vs Dog Image Classifier — Deep Learning (CNN)

2024

Python · TensorFlow · Keras · Computer Vision · Deep Learning

- Designed a Convolutional Neural Network with multiple convolution, max-pooling, and dropout layers using TensorFlow and Keras.
- Applied image preprocessing (resizing, pixel normalization) and data augmentation (flipping, rotation, zoom) to improve model robustness and generalization on unseen images.

AI Job Recommendation System

2025

Python · NLP · Scikit-learn · Recommendation Engine

- Built an ML-powered job recommendation engine matching candidate skill profiles to job roles using cosine similarity and classification models.
- Combined NLP for text feature extraction with supervised learning for role matching; evaluated using precision and recall, iterating to improve match relevance.

Server Monitoring & Performance Data Pipeline

2025

Python · Pandas · Power BI · ETL · Data Engineering

- Designed an end-to-end ETL pipeline to ingest raw server logs, clean and transform metrics, and load results into a structured format for analysis.
- Tracked KPIs (CPU utilization, memory usage, network traffic) with modular, production-style Python scripts and Power BI dashboards.

EDUCATION

Bharathidasan University — M.Sc. Data Science

2024 – 2026

Tiruchirappalli, Tamil Nadu — Relevant coursework: Machine Learning, Deep Learning, Big Data Analytics, Data Mining, NLP, Statistical Computing

St. Joseph's College — B.Sc. Statistics

2021 – 2024

Tiruchirappalli, Tamil Nadu | Grade: 77.32% — Relevant coursework: Probability, Statistical Inference, Operations Research

CERTIFICATIONS

Data Analytics using Python | Power BI – SkillNation | SPSS Certification & Data Mining Workshop | Fundamentals of Python